AHMED PASHA

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 1+ year of hands-on experience designing, training, and deploying production-grade machine learning and deep learning models. Proficient in Python, PyTorch, and Hugging Face Transformers. Experienced in end-to-end ML pipeline development, model evaluation, and optimization. Skilled in computer vision (YOLO, OpenCV, Vision Transformers), LLM application development (LangChain, RAG pipelines), multi-agent AI systems (LangGraph), and MLOps workflows (Docker, Kubernetes, GCP, MLflow).

TECHNICAL SKILLS

Languages:	Python, SQL
ML / DL:	PyTorch, TensorFlow, Keras, Scikit-learn — regression, classification, clustering, ensemble methods
Transformers:	Hugging Face Transformers, GPT, Vision Transformers (ViT)
Computer Vision:	YOLO, OpenCV, image classification, object detection, segmentation (COCO format)
LLM & GenAI:	LangChain, LangGraph, RAG Pipelines, Multi-Agent Systems, OpenAI API, Claude API, Ollama, Llama, Groq, FAISS, ChromaDB
MLOps & Deploy:	Docker, Kubernetes, MLflow, DVC, DagsHub, FastAPI, Flask, Streamlit, GCP, Prometheus, Grafana
Evaluation:	mAP, Precision, Recall, F1-score, confusion matrices, model optimization, hyperparameter tuning
Soft Skills:	Remote collaboration, independent work, agile teamwork, clear technical communication, problem-solving

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Pavaman Technologies — **Jan 2025 – Present | Hyderabad, India**

- Designed and deployed end-to-end ML pipeline combining YOLO + Vision Transformer (ViT) for real-time plant disease detection, solving production challenges like occlusion and variable lighting.
- Built LangChain-based RAG pipelines integrating FAISS vector search and LLMs to generate contextual, document-grounded disease diagnoses and treatment recommendations from image inputs.
- Constructed large-scale annotated training datasets in YOLO and COCO formats; implemented preprocessing, augmentation, and structured data ingestion pipelines for model training.
- Performed systematic model evaluation using mAP, Precision, Recall, and F1-score; conducted error analysis, edge-case validation, and latency optimization for scalable real-time inference.
- Collaborated with cross-functional product and engineering teams, iterating on model performance based on real-world user feedback to deliver a production-ready YOLO + ViT + LLM system.

KEY PROJECTS

Multi AI Agent Research System — *LangGraph + Tavily + FastAPI + Groq + Multiple Models*

- Architected a multi-agent AI system using LangGraph to orchestrate specialized agents.
- Integrated Tavily search API for real-time web retrieval and Groq-accelerated inference with multiple LLMs (Llama, Mixtral) enabling fast, accurate, multi-source responses.
- Exposed agent workflows via a Fast API backend with async endpoints, enabling scalable API-driven access to the multi-agent pipeline for downstream applications.

Medical RAG Assistant — *LangChain + FAISS + Groq + Ocr*

- Built end-to-end Medical RAG system with PDF ingestion (including scanned PDFs via pytesseract OCR), semantic chunking, and HuggingFace sentence-transformer embeddings indexed in FAISS.
- Generated dense vector embeddings using HuggingFace sentence-transformers and indexed them in FAISS for fast similarity search, enabling accurate retrieval of relevant medical context.
- Deployed as an interactive Streamlit application with FastAPI backend, supporting multi-document ingestion and real-time Q&A for healthcare professionals.

RAG-Based AI Assistant — *Llama 3.2 + ChromaDB*

- Developed an end-to-end RAG system with query rewriting, embedding generation, ChromaDB vector search, and reranking; integrated Llama 3.2 via Ollama for accurate document-grounded responses.

EDUCATION

Bachelor of Computer Application (BCA) — ***2021 – 2024 | GPA: 8.2/10***

S.S. Tegnoor / Gulbarga University, Karnataka

CERTIFICATIONS

LLM Engineer | MLOps | Computer Vision | Data Science| Data Science Internship